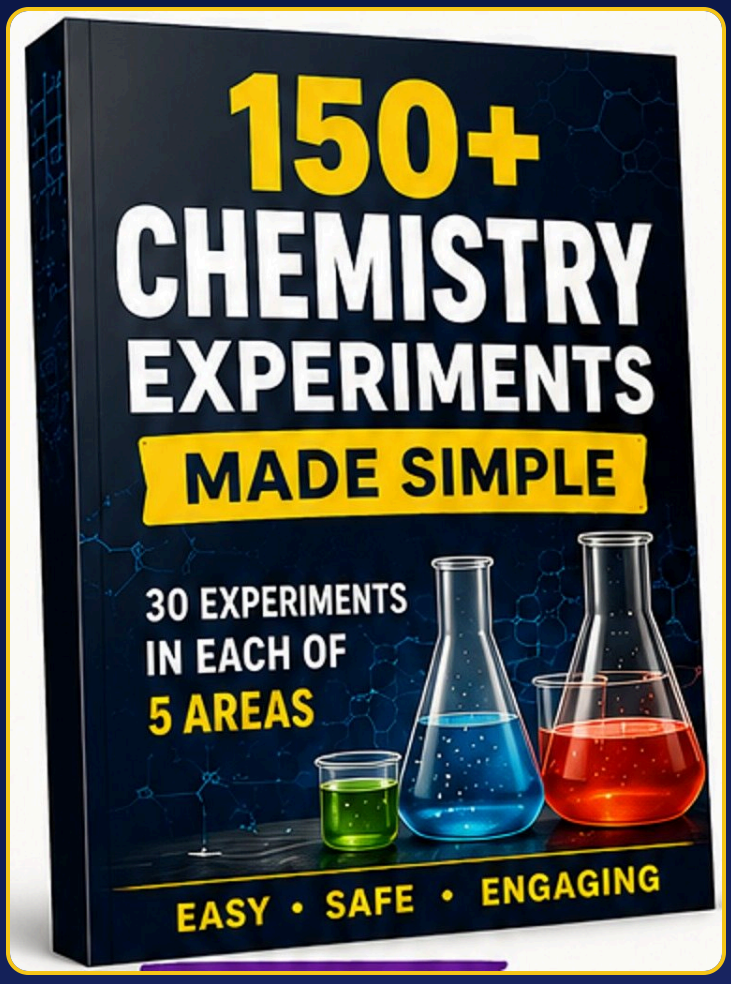
BONUS 4

Year-Long Chemistry Plan (editable)

36 weeks, one experiment a week, mapped to NGSS. PDF + editable Word file



How to use the plan

One experiment a week for 36 weeks, grouped in 6 units. Each week lists the experiment card, the NGSS performance expectations it supports, the prep and class time and a suggested assessment from Bonus 5. The editable Word file has the same table so you can change weeks, add holidays or swap experiments.

> **A simple weekly rhythm**
>
> Monday: preview the question and the safety note. Tuesday or Wednesday: run the experiment. Thursday: record data and write the CER. Friday: exit ticket and discuss.

The 36-week plan

WK	UNIT	EXPERIMENT	NGSS	GRADES	PREP / CLASS	ASSESS WITH	NOTES
1	Unit 1	GEN-02 Liquid Density Tower	MS-PS1-2	6-12	10 / 20-30 min	Observation sheet	
2	Unit 1	GEN-03 Homemade Lava Lamp	MS-PS1-2, HS-PS1-2	6-12	10 / 20-30 min	CER write-up	
3	Unit 1	GEN-04 Oil, Water and Soap	MS-PS1-1, MS-PS1-2	6-12	10 / 20-30 min	Exit ticket	
4	Unit 1	GEN-05 Mixtures vs. Solutions	MS-PS1-2	6-8	10 / 20-40 min	Group work checklist	
5	Unit 1	GEN-07 Marker Chromatography	MS-PS1-2	6-12	10 / 20-30 min	Lab rubric (4-point)	
6	Unit 1	GEN-08 Magnetic Separation	MS-PS1-2	6-8	10 / 20-30 min	Data table and graph check	
7	Unit 1	GEN-13 Physical vs. Chemical Change Stations	MS-PS1-2	6-8	15 / 30-40 min	Observation sheet	
8	Unit 1	GEN-17 Cleaning Pennies	MS-PS1-2, HS-PS1-2	6-12	10 / 20-30 min	CER write-up	
9	Unit 2	ACI-01 Egg Without a Shell	MS-PS1-2, MS-PS1-5	6-12	10 / 20-40 (plus 24-48 h wait) min	Exit ticket	
10	Unit 2	ACI-02 Red Cabbage pH Indicator	MS-PS1-2	6-12	15 / 20-40 min	Group work checklist	
11	Unit 2	ACI-04 Turmeric Test: Soap Turns Red	MS-PS1-2	6-8	10 / 20-30 min	Lab rubric (4-point)	
12	Unit 2	ACI-07 Invisible Ink: Baking Soda and Grape Juice	MS-PS1-2	6-8	10 / 20-30 min	Data table and graph check	
13	Unit 2	ACI-10 Clean the Penny	MS-PS1-2	6-8	10 / 20-30 min	Observation sheet	
14	Unit 2	ACI-12 Nail Rust Race	MS-PS1-2, MS-ETS1-3	6-12	15 / 20-40 (plus 5-day check) min	CER write-up	
15	Unit 3	PHY-02 Ice, Water, and Steam	MS-PS1-4	6-12	15 / 20-30 min	Exit ticket	
16	Unit 3	PHY-03 Salt, Sugar, or Plain? Melting Ice Race	MS-PS1-4, MS-ETS1-3	6-8	10 / 20-40 min	Group work checklist	
17	Unit 3	PHY-05 Cold Pack from Baking Soda and Vinegar	MS-PS1-6, HS-PS1-4	6-12	10 / 20-30 min	Lab rubric (4-point)	
18	Unit 3	PHY-09 Fizz Speed vs Surface Area	HS-PS1-5, MS-PS1-2	6-12	10 / 20-30 min	Data table and graph check	

WK	UNIT	EXPERIMENT	NGSS	GRADES	PREP / CLASS	ASSESS WITH	NOTES
19	Unit 3	PHY-13 Which Evaporates Faster? Surface Area Test	MS-PS1-4, MS-ESS2-4	6-8	10 / 20-30 min	Observation sheet	
20	Unit 3	PHY-14 Rain on a Can: Condensation	MS-ESS2-4, MS-PS1-4	6-8	10 / 20-30 min	CER write-up	
21	Unit 4	ORG-02 Pineapple vs Gelatin: Enzymes at Work	MS-LS1-7, MS-PS1-2	6-12	15 / 20-30 min	Exit ticket	
22	Unit 4	ORG-05 Yeast Balloon Inflator	MS-LS1-7, MS-PS1-2	6-8	10 / 20-30 min	Group work checklist	
23	Unit 4	ORG-07 Curds and Whey: Cheese Chemistry	MS-PS1-2, MS-LS1-7	6-8	15 / 30-40 min	Lab rubric (4-point)	
24	Unit 4	ORG-09 Polymer Slime	MS-PS1-3, MS-PS1-2	6-8	10 / 20-30 min	Data table and graph check	
25	Unit 4	ORG-10 Oobleck: Liquid or Solid?	MS-PS1-1, MS-PS1-2	6-8	5 / 20-30 min	Observation sheet	
26	Unit 4	ORG-14 Butter in a Jar	MS-PS1-2, MS-LS1-7	6-8	5 / 20-30 min	CER write-up	
27	Unit 5	ENV-02 Acid Rain vs Chalk	MS-PS1-2, MS-ESS3-3	6-8	10 / 20-40 min	Exit ticket	
28	Unit 5	ENV-04 Acid Rain and a Leaf	MS-ESS3-3, MS-PS1-2	6-8	10 / 20-40 min	Group work checklist	
29	Unit 5	ENV-07 Why Fizzy Water Goes Flat	MS-PS1-4, MS-PS1-2	6-8	10 / 20-40 min	Lab rubric (4-point)	
30	Unit 5	ENV-11 Decomposition in Compost Jars	MS-LS1-7, MS-ESS3-3	6-8	15 / 20-40 min	Data table and graph check	
31	Unit 5	ENV-12 Soil pH with Cabbage Indicator	MS-PS1-2, HS-PS1-2	6-12	15 / 20-40 min	Observation sheet	
32	Unit 5	ENV-13 Soil Filtering Race	MS-ESS2-4, MS-ETS1-3	6-8	10 / 20-40 min	CER write-up	
33	Unit 6	GEN-18 Magic Milk	MS-PS1-2	6-8	5 / 15-30 min	Exit ticket	
34	Unit 6	GEN-19 Diffusion in Hot vs. Cold Water	MS-PS1-4, MS-PS3-4	6-12	10 / 20-30 min	Group work checklist	
35	Unit 6	GEN-20 Walking Water	MS-PS1-1	6-8	10 / 20-40 min	Lab rubric (4-point)	
36	Unit 6	GEN-22 Floating Egg	MS-PS1-2, HS-PS1-3	6-12	10 / 20-30 min	Data table and graph check	

Unit overview

Unit 1: Matter, mixtures and solutions

8 weeks. NGSS performance expectations supported: HS-PS1-2, MS-PS1-1, MS-PS1-2.

- GEN-02: Liquid Density Tower (Easy)
- GEN-03: Homemade Lava Lamp (Easy)
- GEN-04: Oil, Water and Soap (Easy)
- GEN-05: Mixtures vs. Solutions (Easy)
- GEN-07: Marker Chromatography (Easy)

- GEN-08: Magnetic Separation (Easy)
- GEN-13: Physical vs. Chemical Change Stations (Easy)
- GEN-17: Cleaning Pennies (Easy)

Unit 2: Acids, bases and metals

6 weeks. NGSS performance expectations supported: MS-ETS1-3, MS-PS1-2, MS-PS1-5.

- ACI-01: Egg Without a Shell (Easy)
- ACI-02: Red Cabbage pH Indicator (Easy)
- ACI-04: Turmeric Test: Soap Turns Red (Easy)
- ACI-07: Invisible Ink: Baking Soda and Grape Juice (Easy)
- ACI-10: Clean the Penny (Easy)
- ACI-12: Nail Rust Race (Easy)

Unit 3: Energy, rates and gases

6 weeks. NGSS performance expectations supported: HS-PS1-4, HS-PS1-5, MS-ESS2-4, MS-ETS1-3, MS-PS1-2, MS-PS1-4, MS-PS1-6.

- PHY-02: Ice, Water, and Steam (Easy)
- PHY-03: Salt, Sugar, or Plain? Melting Ice Race (Easy)
- PHY-05: Cold Pack from Baking Soda and Vinegar (Easy)
- PHY-09: Fizz Speed vs Surface Area (Easy)
- PHY-13: Which Evaporates Faster? Surface Area Test (Easy)
- PHY-14: Rain on a Can: Condensation (Easy)

Unit 4: The chemistry of living things

6 weeks. NGSS performance expectations supported: MS-LS1-7, MS-PS1-1, MS-PS1-2, MS-PS1-3.

- ORG-02: Pineapple vs Gelatin: Enzymes at Work (Easy)
- ORG-05: Yeast Balloon Inflator (Easy)
- ORG-07: Curds and Whey: Cheese Chemistry (Easy)
- ORG-09: Polymer Slime (Easy)
- ORG-10: Oobleck: Liquid or Solid? (Easy)
- ORG-14: Butter in a Jar (Easy)

Unit 5: Chemistry and the environment

6 weeks. NGSS performance expectations supported: HS-PS1-2, MS-ESS2-4, MS-ESS3-3, MS-ETS1-3, MS-LS1-7, MS-PS1-2, MS-PS1-4.

- ENV-02: Acid Rain vs Chalk (Easy)
- ENV-04: Acid Rain and a Leaf (Easy)

- ENV-07: Why Fizzy Water Goes Flat (Easy)
- ENV-11: Decomposition in Compost Jars (Easy)
- ENV-12: Soil pH with Cabbage Indicator (Easy)
- ENV-13: Soil Filtering Race (Easy)

Unit 6: Review and choice weeks

4 weeks. NGSS performance expectations supported: HS-PS1-3, MS-PS1-1, MS-PS1-2, MS-PS1-4, MS-PS3-4.

- GEN-18: Magic Milk (Easy)
- GEN-19: Diffusion in Hot vs. Cold Water (Easy)
- GEN-20: Walking Water (Easy)
- GEN-22: Floating Egg (Easy)

Pacing notes

- Plan a buffer week before long breaks. Replace any week with another card of the same area.
- Multi-day experiments (crystals, rusting, mold, compost) can start on Monday and be observed all week.
- Gather the materials for a whole unit at once; most are grocery or pharmacy items.
- Align this plan with your district pacing guide and the standards your state uses.

Next step

Open the editable Word file, add your school calendar, and mark week 1.